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Machine learning-based prediction for incidence of endoscopic retrograde cholangiopancreatography after emergency laparoscopic cholecystectomy: A retrospective, multicenter cohort study

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Abstract

Background Laparoscopic cholecystectomy is the preferred treatment for symptomatic cholelithiasis and acute cholecystitis, with increasing applications even in severe cases. However, the possibility of postoperative endoscopic retrograde cholangiopancreatography (ERCP) to manage choledocholithiasis or biliary injuries poses significant clinical challenges. This study aimed to develop a predictive model for ERCP incidence following emergency laparoscopic cholecystectomy using advanced machine learning techniques.

Methods We conducted a retrospective cohort study using the Tokushukai Medical Database, which includes data from 42 hospitals. The study population consisted of adult patients undergoing emergency laparoscopic cholecystectomy. We used four machine learning models—logistic regression, random forest, gradient-boosting decision trees (GBDTs), and multilayer perceptrons on a dataset divided into training/validation and testing groups. We also calculated Shapley additive explanation values for GBDTs to identify variables with larger feature importance.

Results Of 9,695 patients from July 2010 to June 2020, 8,854 met the inclusion criteria. The incidence of postoperative ERCP was 5.7% (362/6,377) and 6.4% (158/2477) in the training/validation and testing datasets, respectively. The GBDT demonstrated superior performance, with the highest predictive capacity for postoperative ERCP. Significant predictors identified included common bile duct dilatation on CT or ultrasound, serum albumin, and lactate dehydrogenase levels, which showed larger feature importance.

Conclusion This study successfully developed a robust predictive model for ERCP following emergency laparoscopic cholecystectomy.

Keywords Laparoscopic cholecystectomy · Endoscopic retrograde cholangiopancreatography · Machine learning · Prediction

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Laparoscopic cholecystectomy is currently the standard of care for gallbladder removal and is performed to manage symptomatic cholelithiasis, including emergency intervention for acute cholecystitis. Initially only indicated for those with mild to moderate severity, updated Tokyo Guidelines 2018 extended the indication of laparoscopic cholecystectomy to cholecystitis with any severity provided the appropriate surgeon skillset and the facility to accommodate the burden of perioperative care [1–3]. However, serious complications, including bleeding, abscess formation, and iatrogenic bile duct injuries, have been reported in 2.6% of patients [4], and approximately, 8–20% of patients with cholelithiasis experience concurrent choledocholithiasis [5,

6], where endoscopic retrograde cholangiopancreatography (ERCP) or intraoperative cholangiography (IOC) is needed.

ERCP is an advanced endoscopic procedure wherein a specialized side-viewing upper endoscope is guided into the duodenum, cannulating the ampulla of Vater to access the biliary and pancreatic ducts. A contrast medium injection opacifies the ducts, allowing radiographic visualization and therapeutic interventions. ERCP is one of the most common procedures for choledocholithiasis or iatrogenic biliary injury because of its high sensitivity and specificity while allowing simultaneous therapeutic intervention. However, overall rates of ERCP-related adverse events, including pancreatitis, perforation, bleeding, and infection, ranged from 5 to 12%, and mortality rates ranged from 0.1 to 1.4% [7–9].

IOC is another modality to confirm choledocholithiasis or define biliary anatomy, allowing on-table progression to therapeutic laparoscopic common bile duct exploration (LCBDE). Early detection and treatment of these conditions, including iatrogenic biliary injuries, can prevent further progression of the injury [10]. Some studies, however, reported that IOC is associated with increased cost, operation time, and length of hospital stay, with no substantial evidence indicating that it improves stone extraction and biliary injury rates [11]. When IOC was routinely performed, 10% of patients had asymptomatic common bile duct (CBD) stones, many of which may subsequently pass spontaneously or never cause any symptoms [12, 13].

In the emergency setting, surgeons may face challenges on whether to prioritize the intervention for acute cholecystitis complicated with choledocholithiasis or to predict the perioperative morbidity risk associated with ERCP. In this study, we developed machine learning models with nonlinear feature extraction to create a precise prediction model for the incidence of postoperative ERCP in patients who underwent emergency laparoscopic cholecystectomy. We achieved this using clinical data at admission and a conventional logistic regression model that has frequently been applied to linear feature extraction was used for analysis. We also attempted to identify predictive factors associated with the incidence of postoperative ERCP.

Materials and methods

Data source

We used the Tokushukai Medical Database for this retrospective cohort study, generated from the clinical data from 42 hospitals in the Tokushukai Group [14, 15]. The Tokushukai Group is one of the largest Japanese hospital chains, which handles 72 hospitals, including 38 hospitals utilizing the diagnosis procedure combination (DPC) system in Japan [16]. The Tokushukai Medical Database mainly includes departmental claims data (particularly the DPC inpatient data) in addition to electronic medical notes, including outpatient and inpatient laboratory findings. The DPC inpatient data consist of patients' sex and age; dates of admission and discharge; discharge status (alive or dead); primary diagnoses, pre-existing comorbidities on admission, and complications during hospital stay documented by the attending doctors based on the International Classification Disease 10th revision (ICD-10) codes [17]; types of surgery; and medications and interventions, including renal replacement therapy and mechanical ventilation.

The Tokushukai Group Joint Ethics Committee approved our study following the tenets of the Declaration of Helsinki. The requirement for informed consent from patients was deferred, because all data were anonymized for research purposes.

Study population

Our study population included adult patients (aged ≥ 18 years) who were admitted to one of the 42 hospitals engaging in the Tokushukai Medical Database between July 2010 and June 2020 and who underwent emergency laparoscopic cholecystectomy. Emergency surgery was defined as surgery performed within two days from admission.

The patients were divided into two groups: Those hospitalized between July 2010 and December 2017 (training/validation dataset) and between January 2018 and June 2020 (testing dataset). The patient characteristics (i.e., candidate predictors) and in-hospital mortality were described using summary statistics.

Machine learning

We applied the following four established machine learning analyses to the training/validation dataset to generate prediction models: Logistic regression (LR) [18], random forest (RF) [19], GBDTs [20], and multilayer perceptrons (MLPs) [21]. Although some studies applied machine learning analyses to prediction models for in-hospital mortality [15, 22], we used the incidence of ERCP after laparoscopic cholecystectomy as the objective variable in this study. As predictor variables, 53 variables were used for model training, including baseline characteristics, such as age, sex, weight, height, body mass index (BMI) upon admission, length of stay to surgery, primary diagnosis, preceding procedures, use of IOC, CBD dilatation (> 6 mm) on CT or ultrasound, 15 pre-existing comorbidities, and 20 laboratory variables on admission. In addition, we applied a class-balanced loss to the training phase of machine learning, and all imputed data were used to develop the model and adopt the hyperparameters of the model, which can magnify the mean values of the area under the receiver operating characteristics curve

(AUROC) obtained from fivefold cross-validation within the training data. The training data were further divided into training and validation datasets.

In the LR, we applied a grid search to regulate the coefficient hyperparameters. In the RF, we also used a grid search to utilize the model with 500 decision trees for the maximum depth of each decision tree and explanatory variable applied to classification in the node arrangement. For the GBDT, we investigated the number of leaves and the required data in each leaf with a grid search to develop the model. For MLP, we applied a four-layer neural network with each layer of 256 perceptrons to the stochastic gradient descent. The learning rate, dropout ratio of perceptrons, and batch size were tested with a grid search to optimize the hyperparameters. The validation data were used to assess the termination mark of the training with the early stopping protocol in the multilayer perceptron analysis. Python (version 3.7.7) and R (version 4.1.3) scripts were used for machine learning analyses.

Following the training and hyperparameter optimization, the prediction models were processed using all testing datasets, and the AUROC was calculated for each model. Regarding the output value of the machine learning model, isotonic regression was applied to the probability calibration with the validation data at each cross-validation phase to evaluate the predictive capacity for future data. Therefore, the mean output values of these models created with cross-validation were used to calculate the concluding AUROC. We also calculated the area under the precision-recall curve (AUPRC) for each model as per the prediction assignment with imbalanced data. Regarding a precise comparison of prediction analyses in binary classification with imbalanced data, AUPRC can be more descriptive than AUROC [23]. The Delong test was used to compare the AUROC and AUPRC, with LR as the reference [24]. Shapley additive explanations (SHAP) values were also calculated for the GBDT to identify the significant variables with the SHAP module of Python [25].

Results

Between July 2010 and June 2020, 9,695 patients who underwent emergency laparoscopic cholecystectomy were identified in the database. We excluded 841 patients because they did not undergo surgery within 2 days after admission ($n = 207$) and/or had missing variables ($n = 634$). We included the remaining 8,854 patients, including 6,377 and 2,477 patients in the training/validation and testing datasets, respectively (Fig. 1). Table 1

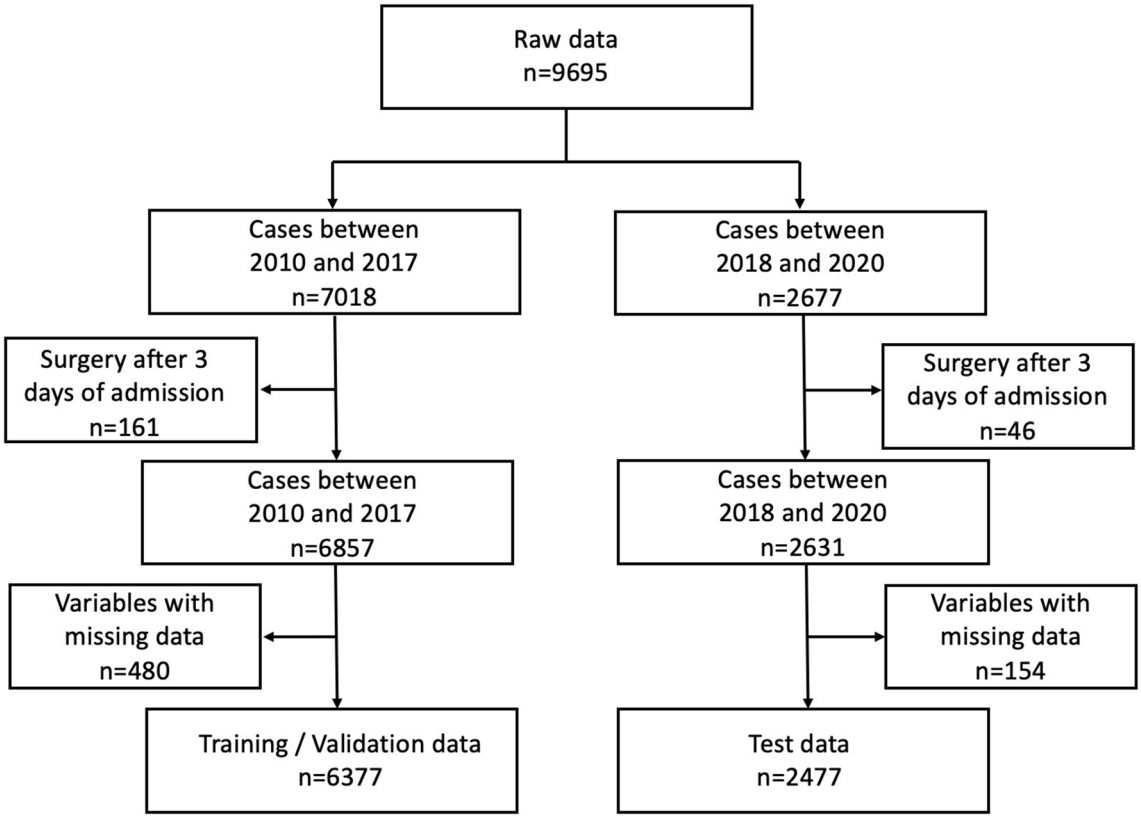


Fig. 1 Schema of selection of the study participants and data processing

Table 1 Baseline characteristics and statistic variables of the study participants with emergency laparoscopic cholecystectomy (training/validation versus testing)

Baseline variables and in-hospital factors	Training/Validation	Testing	P value
	n = 6377	*n* = 2477	
Post-operative ERCP	362 (5.7%)	158 (6.4%)	
Baseline variables and in-hospital factors			
Age	65.01 ± 15.58	66.30 ± 15.90	0.21
Sex (%)			
Female	2630 (41.1%)	1009 (41.0%)	
Male	3747 (58.9%)	1468 (59.0%)	
Weight (kg), mean (SD)	63.13 ± 13.74	64.42 ± 14.65	0.34
Height (cm), mean (SD)	160.85 ± 9.45	161.11 ± 9.46	0.75
BMI (kg/m2), mean (SD)	24.26 ± 4.19	24.66 ± 4.39	0.87
Length of stay in hospital, days, mean (SD)	20.36 ± 38.90	16.05 ± 21.70	< 0.01
Length of stay to surgery	2.54 ± 1.65	2.85 ± 1.15	0.03
Primary diagnosis			
Calculous cholecystitis	5956 (93.4%)	2296 (92.7%)	
Acalculous cholecystitis	179 (2.8%)	69 (2.7%)	
Malignant HPB tumor	47 (0.7%)	23 (0.9%)	
Benign HPB tumor	99 (1.6%)	33 (1.3%)	
Others	96 (1.5%)	56 (2.3%)	
CT/Ultrasound finding			
Biliary dilatation (> 7 mm)	1457 (22.8%)	469 (19.0%)	
Preceding intervention			
ERCP	1217 (19.1%)	409 (16.5%)	
Cholecystostomy	536 (8.4%)	170 (6.9%)	
Intraoperative intervention			
IOC	882 (13.8%)	294 (11.9%)	
Pre-existing comorbidity			
Sepsis	119 (1.9%)	41 (1.7%)	
Dehydration	617 (9.7%)	201 (8.1%)	
Malignancy	245 (3.8%)	121 (4.9%)	
Hypertension	1797 (28.2%)	826 (33.3%)	
Dyslipidemia	836 (13.1%)	468 (18.9%)	
Diabetes	1084 (17.0%)	453 (18.3%)	
COPD	87 (1.4%)	24 (1.0%)	
Heart failure	189 (3.0%)	65 (2.6%)	
Ischemic heart disease	445 (7.0%)	173 (7.0%)	
Atrial fibrillation	180 (2.9%)	124 (5.0%)	
Cerebrovascular disease	443 (6.9%)	184 (7.4%)	
Liver disease	364 (5.7%)	114 (4.6%)	
Chronic kidney disease	145 (2.3%)	54 (2.2%)	
Dimentia	174 (2.7%)	124 (5.0%)	
Psychological disease	239 (3.7%)	114 (4.6%)	
Blood test at admission, mean (SD)			
White blood cell count (10^9/l), mean (SD)	11.13 ± 4.75	11.43 ± 4.90	0.68
Hemoglobin (g/dl), mean (SD)	13.71 ± 2.03	13.73 ± 2.02	0.80
Platelet count (10^9/l), mean (SD)	220.59 ± 76.09	226.62 ± 77.20	0.22
Sodium (mmol/l), mean (SD)	138.63 ± 3.48	138.49 ± 3.31	0.88
Potassium(mmol/l), mean (SD)	3.98 ± 0.49	3.96 ± 0.44	0.76
Chlorine (mmol/l), mean (SD)	102.25 ± 3.94	102.46 ± 3.70	0.72
Creatinine (mg/dl), mean (SD)	0.94 ± 0.90	0.94 ± 0.81	0.87

Table 1 (continued)

Baseline variables and in-hospital factors	Training/Validation	Testing	P value
Blood urea nitrogen (mg/dl), mean (SD)	16.92 ± 10.07	17.02 ± 10.41	0.34
Total protein (g/dl), mean (SD)	7.02 ± 0.66	7.00 ± 0.67	0.71
Albumin (g/dl), mean (SD)	4.01 ± 0.67	3.90 ± 0.59	0.23
Alanine aminotransferase (U/l), mean (SD)	88.51 ± 157.95	78.01 ± 138.62	< 0.01
Aspartate transaminase (U/l), mean (SD)	102.39 ± 200.79	90.29 ± 181.59	< 0.01
Total bilirubin (mg/dl), mean (SD)	1.47 ± 1.51	1.44 ± 1.40	0.53
Alkaline phosphatase (U/l), mean (SD)	358.86 ± 276.98	348.24 ± 304.17	0.33
Gamma-glutamyltransferase (U/l), mean (SD)	144.36 ± 220.98	126.46 ± 192.35	< 0.01
Lactate dehydrogenase (U/l), mean (SD)	358.86 ± 276.98	260.19 ± 153.72	< 0.01
Amylase (U/l), mean (SD)	127.13 ± 351.90	99.95 ± 223.13	< 0.01
C-reactive protein (mg/dl), mean (SD)	5.72 ± 8.32	6.11 ± 8.72	0.07
Prothrombin time-International normalized ratio, mean (SD)	1.11 ± 0.23	1.08 ± 0.25	0.34
Partial thromboplastin time (second), mean (SD)	28.75 ± 5.44	29.39 ± 4.09	0.23

shows the detailed statistics of the variables in the preprocessed training/validation and testing datasets.

The incidence of postoperative ERCP was 5.7% (362/6,377) and 6.4% (158/2477) for the training/validation and testing datasets, respectively. Following the model training, the AUROC and AUPRC were obtained based on the results of the analyses using 53 variables with developed machine learning models against the testing dataset.

Accordingly, the AUROC values of 0.712 [95% confidence interval [CI], 0.689–0.731], 0.837 (95% CI, 0.808–0.862; $p < 0.01$), 0.847 (95% CI, 0.821–0.874; $p < 0.01$), and 0.677 (95% CI, 0.645–0.702; $p = 0.07$) and AUPRC values of 0.132 (95% CI, 0.101–0.163), 0.289 (95% CI, 0.254–0.318; $p < 0.01$), 0.307 (95% CI, 0.281–0.345; $p < 0.01$), and 0.110 (95% CI, 0.091–0.139; $p = 0.33$) were calculated for the LR, RF, GBDT, and MLP, respectively (Fig. 2a,b). Consequently, the prediction model generated with the GBDT demonstrated the best AUROC and AUPRC values; otherwise, other models showed a modest discrepancy in the values.

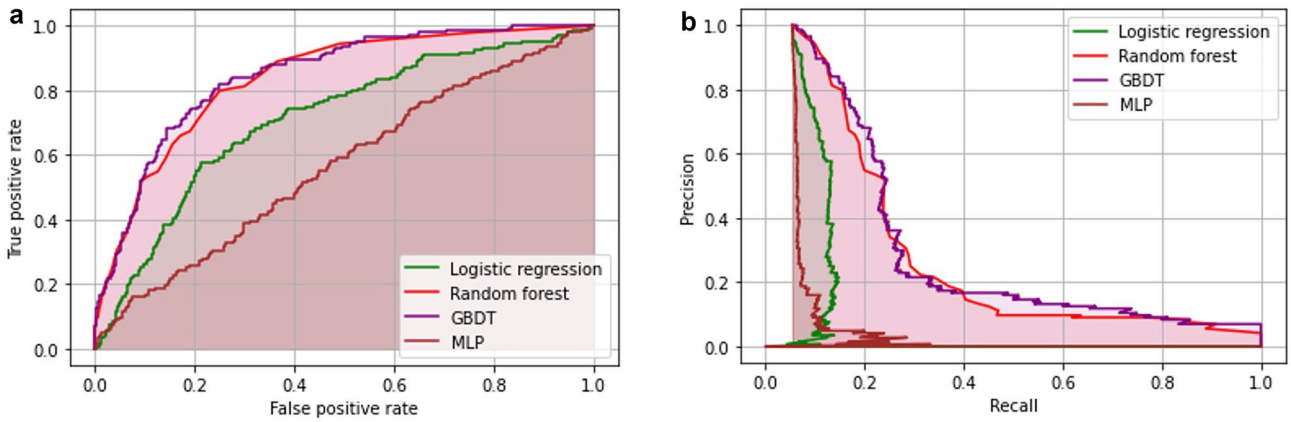
We also calculated SHAP values for the GBDT to scrutinize the relevance of the variables and the validity of the prediction model with the internal algorithm. Biliary dilatation and several laboratory variables, such as serum albumin and lactate dehydrogenase (LDH), were considered high (Fig. 3a). SHAP value distribution was also calculated with the test dataset. Each plot indicates one prediction result in the test dataset. The plot color shows variable values as indicated in color scale bars. A positive SHAP value in the red plot contributes to the outcome. By contrast, a negative SHAP value, indicated by the blue plot, implies the opposite (Fig. 3b). Therefore, biliary dilatation, low serum albumin, and high serum LDH were significantly associated with the incidence of postoperative ERCP.

Discussion

Emergency laparoscopic cholecystectomy is often performed without preoperative imaging of the biliary tree, such as MRCP. In contrast to elective cases, where such imaging is more readily accessible, emergency cholecystectomy, among other confounding factors, may lead to an increased incidence of unanticipated choledocholithiasis and complications, such as iatrogenic bile duct injuries [26, 27]. ERCP is a reasonable and efficient option for addressing these conditions, which can satisfy both diagnostic and therapeutic measures. However, ERCP has inherent risks, including but not limited to pancreatitis, cholangitis, and bleeding, which could be increased with endoscopic sphincterotomy (EST), depending on the complexity of the procedure [28]. Furthermore, a potential disadvantage of postoperative ERCP would require a second anesthesia and potential reoperation for duct clearance if this fails.

IOC is an intraoperative option to define individual biliary anatomy, identify occult bile duct stones, and detect iatrogenic biliary complications, which allows escalation to LCBDE as a therapeutic maneuver. LCBDE has advantages over ERCP, including direct stone removal and stent placement in the CBD without sphincterotomy. It can also be a better option for patients with anatomic variations that make ERCP difficult or risky. Whether IOC should be performed routinely or selectively remains controversial. Indications for IOC during laparoscopic cholecystectomy generally include a history of gallstone pancreatitis, elevated liver enzymes, and CBD > 5–7 mm in diameter [29, 30]. However, the prevalence of IOC or LCBDE is distinctive depending on local expertise and surgeon preference.

In this study, we successfully introduced the machine learning prediction model to determine the incidence of ERCP after emergency laparoscopic cholecystectomy



	AUROC (95% CI)	*P* value
Logistic regression	0.712 (0.689-0.731)	reference
Random forest	0.837 (0.808-0.862)	<0.01
GBDT	0.847 (0.821-0.874)	<0.01
MLP	0.677 (0.645-0.702)	0.07

	AUPRC (95% CI)	*P* value
Logistic regression	0.132 (0.101-0.163)	reference
Random forest	0.289 (0.254-0.318)	<0.01
GBDT	0.307 (0.281-0.345)	<0.01
MLP	0.110 (0.091-0.139)	0.33

Fig. 2 **a** The receiver operating characteristics curve and area under the receiver operating characteristic (AUROC) curve of the models are shown. The results of the prediction using all testing data are described. **b** The precision–recall curve and area under the precision–recall (AUPRC) curve of the models are shown. The results of the prediction using all testing data are described. Both are demonstrated with a 95% confidence interval (CI) and compared using the Delong test with the logistic regression as the reference to calculate the *P* value

with an extensive database, including approximately 9,000 patients in 42 community hospitals in Japan. In the analyses, we used the 52 variables on admission to generate a prediction model using the machine learning models, including LR, RF, GBDT, and MLP. Among these models, GBDT demonstrated the most significant performance for both AUROC and AUPRC. Therefore, we also assessed SHAP values using GBDT results to quantify and visualize their significance. It demonstrated that CBD dilatation is the most robust predictor for ERCP incidence, which was consistent with the current indication for IOC [31, 32]. Additionally, low serum albumin and total protein levels were also relevant factors to the outcome. Although their direct role in cholelithiasis or choledocholithiasis is not widely recognized, they may serve as an indicator of underlying liver conditions, wherein some studies demonstrated that cholelithiasis was associated with nonalcoholic fatty liver disease [31, 32]. Serum LDH was also identified as a significant variable associated with the need for postoperative ERCP. This was, however, considered unspecific because its levels would be deranged not only with inflammation in the biliary system but also with a broad spectrum of tissue damage, including hemolysis. Although liver enzymes, such as alkaline phosphate or gamma-glutamyl transferase, are elevated with cholangitis, this study did not show a significant relationship between preoperative deranged liver enzymes and ERCP incidence after emergency laparoscopic cholecystectomy.

Despite the recent focus on deep learning models like MLPs, they do not always outperform other machine learning techniques in clinical prediction tasks. Here, the GBDT model demonstrated superior predictive performance over the deep learning-based MLP for predicting the need for postoperative ERCP following laparoscopic cholecystectomy. This result underscores that model effectiveness depends more on alignment with the specific data characteristics and clinical prediction goals than on model complexity or the use of deep learning architectures. Therefore, selecting the most appropriate model should be driven by clinical context and the nature of the dataset rather than a general preference for advanced model architecture.

This study has several limitations. First, the dataset does not include imaging or indications for laparoscopic cholecystectomy. We expect that the treatment indications in this dataset were homogeneous because most Japanese hospitals manage their patients with cholecystitis or cholangitis following the Tokyo Guideline 2018 [1]. However, variance might occur depending on imaging findings, such as CT or MRCP, where we can potentially change the strategy. The dataset in this study does not include the indication for postoperative ERCP (e.g., postoperative

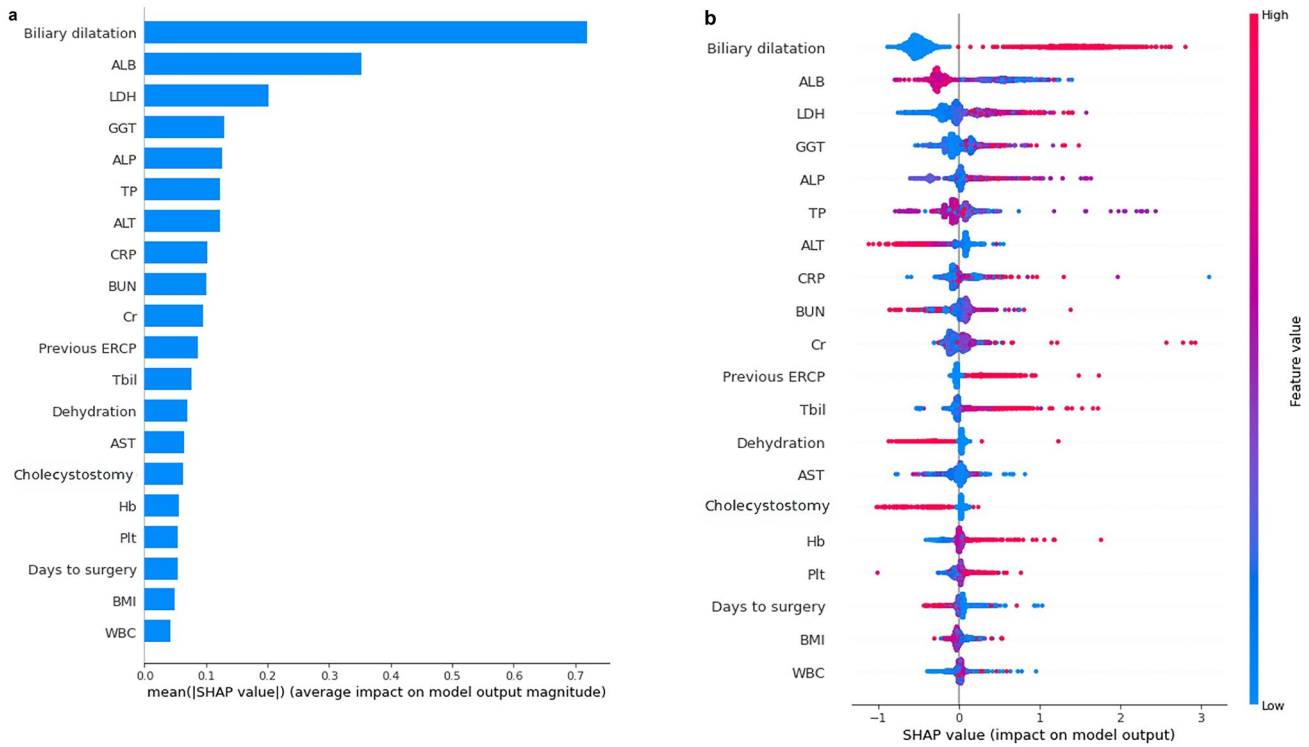


Fig. 3 The Shapley additive explanation (SHAP) values of the top 20 variables based on the result of the gradient-boosting trees are demonstrated in order of the mean absolute SHAP value. **a** The bar graph shows the mean total SHAP value in each variable. **b** The SHAP value distribution was also calculated with the testing dataset. Each plot indicates one prediction result in the testing dataset. The plot colors show variable values indicated in color scale bars. A positive SHAP value in the red plot contributes to in-hospital mortality. A negative SHAP value, indicated by the blue plot, implies the opposite

CBD stone or iatrogenic biliary injury) and the detailed procedure during the ERCP (e.g., sphincterotomy, balloon dilatation, choledocholithotomy, or stent insertion). Second, while GBDT in this study demonstrated high predictive accuracy, GBDT and MLP are generally susceptible to overfitting and is sensitive to outliers, which could lead to reduced generalizability and potentially misleading predictions in diverse patient populations, particularly in atypical cases.

For the clinical implications, while the model achieved an AUROC of 0.847, which indicates strong predictive performance, this study focused on risk stratification where surgeons could consider other perioperative investigations, such as MRCP or IOC, for patients considered high risk so that they can minimize the incidence of postoperative ERCP. Although our analysis results might be coherent with the prediction models with internal algorithms in existing studies, it does not necessarily justify the cause-and-effect relationship between variables and outcomes. Therefore, in actual clinical situations, prioritizing risk assessment with existing prediction models that are clinically established and reassessing patients in the perioperative period is essential.

Conclusion

This study successfully developed a robust predictive model for ERCP following emergency laparoscopic cholecystectomy. Our findings can assist clinicians in the identification of high-risk patients for targeted management, potentially reducing the need for postoperative ERCP and its associated risks. Future studies should focus on integrating real-time imaging data and expanding patient demographics to enhance the model's applicability and accuracy.

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Declarations

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